

11.2 Fundamental Particles

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.2 Fundamental Particles
Difficulty	Hard

Time allowed: 10

Score: /7

Percentage: /100

Question 1

Quarks are thought to make up protons and neutrons.

The 'up' quark has a charge of $\frac{2}{3}e$; a 'down' quark has a charge of $-\frac{1}{3}e$, where e is the elementary charge ($+1.6 \times 10^{-19} \text{ C}$)

How many up quarks and down quarks must a proton contain?

	Up quarks	Down quarks
A	0	3
B	2	1
C	1	2
D	1	1

[1 mark]

Question 2

A CERN detector shows a signal only in the hadronic calorimeter. No signal is observed in the tracker, electromagnetic calorimeter or muon chambers.

Which particle must be responsible for this signal?

- A. pion
- B. neutrino
- C. photon
- D. neutron

[1 mark]

Question 3

Some particles are made up of a combination of three quarks.

Which particle would **not** give a particle a charge of either $+1.6 \times 10^{-19} \text{ C}$ or zero?

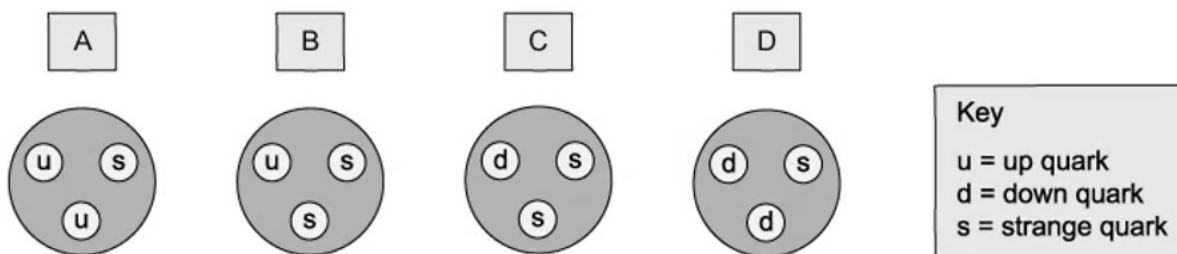
- A. up, up, up
- B. up, up, down
- C. up, strange, strange
- D. up, down, down

[1 mark]

Question 4

The composition of four different hadrons are shown in the diagram below. One of the hadrons is a Σ^+ particle. It has a charge of $+e$, where e is the elementary charge.

Which hadron could be the Σ^+ particle?



[1 mark]

Question 5

When high-speed hadrons collide in particle accelerators, a positive pion (π^+) is an unstable particle produced. The π^+ particle has a charge of $+e$.

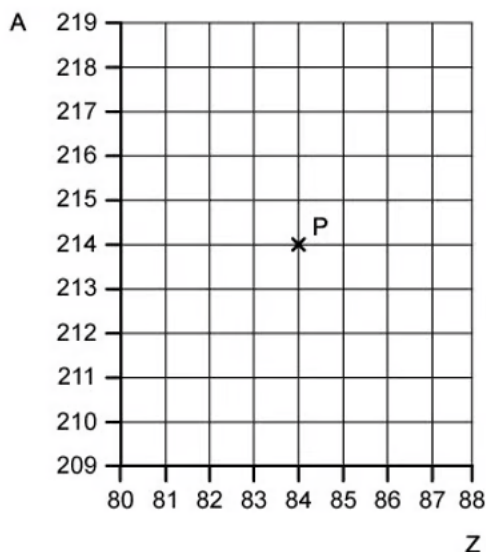
What is the quark combination of the π^+ particle?

- A. $u \bar{u}$
- B. $u \bar{d}$
- C. $d \bar{u}$
- D. $d \bar{d}$

[1 mark]

Question 6

A graph of nucleon number A against proton number Z is shown in the diagram below.



The graph shows a cross that represents a nucleus P.

Nucleus P decays by emitting an α -particle to form a nucleus Q.

Nucleus Q then decays by emitting a β^- particle to form a nucleus R.

The quark composition of one nucleon in Q is changed during the emission of the β^- particle.

What is the corresponding change in quark composition from nucleus Q to nucleus R?

- A. no change
- B. u -quark to d -quark
- C. d -quark to u -quark
- D. d -quark to s -quark

[1 mark]

Question 7

The equation for β^- decay is

$$n \rightarrow p + e^- + \bar{\nu}_e$$

Using your knowledge of antiparticles and the weak interaction, what is the equation for β^+ decay?

A. $\bar{n} \rightarrow \bar{p} + e^+ + \nu_e$

B. $p \rightarrow n + e^+ + \nu_e$

C. $n + e^+ \rightarrow p + \bar{\nu}_e$

D. $p \rightarrow n + e^+ + \bar{\nu}_e$

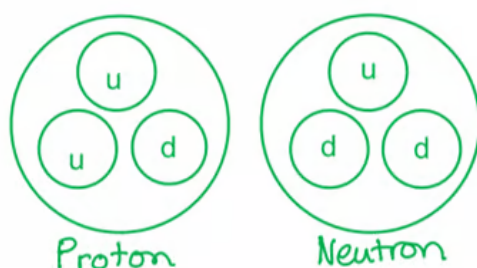
[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- Protons and neutrons are baryons composed of up and down quarks held together by gluons
- They are both made of three quarks as shown in the diagram



- A proton consists of two up quarks and one down quark,
 - $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +e$
- This gives an overall charge of $+e$, therefore, a proton has a quark combination of *uud*

2

The correct answer is **D** because:

- The hadronic calorimeter measures the energy of hadrons
 - Hadrons are particles made of quarks such as baryons or mesons
- No electromagnetic signal was detected, so, the particle must not have a charge
- The only hadron in the list is a neutron; a neutron is made up of two down quarks and one up quark and is neutrally charged overall
 - Therefore, the detected particle must be a neutron

A is incorrect because a pion is a type of meson; a meson contains one quark and one anti-quark bound by a gluon

B is incorrect because neutrinos are leptons with no charge

C is incorrect because photons are the smallest discrete amount or quantum of electromagnetic radiation; it is the basic unit of all light

3

The correct answer is **A** because:

- The 'up' quark has a charge of $\frac{2}{3}e$
- The 'down' quark has a charge of $-\frac{1}{3}e$
 - Where e is the elementary charge, $+1.6 \times 10^{-19} \text{ C}$
- The 'strange' quark has a charge of $-\frac{1}{3}e$
- Consider option **A**:
 - $\frac{2}{3}e + \frac{2}{3}e + \frac{2}{3}e = +2e$
- This is correct as it not neutral as it is double $+1.6 \times 10^{-19} \text{ C}$

B is incorrect
because

$$\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +e$$

a quark combination of up, up down is the combination in a proton and would have a charge of $+1.6 \times 10^{-19} \text{ C}$

C is incorrect
because

$$\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$$

a quark combination of up, strange, strange would give a charge of zero

D is incorrect
because

$$\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$$

a quark combination of up, down, down would give a charge of

zero

4

The correct answer is **A** because:

- Hadrons are particles made of quarks such as baryons or mesons
- The 'up' quark has a charge of $\frac{2}{3}e$
- The 'down' quark has a charge of $-\frac{1}{3}e$
- The 'strange' quark has a charge of $-\frac{1}{3}e$
 - Where e is the elementary charge, $+1.6 \times 10^{-19} \text{ C}$
- Consider option **A**:
 - $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +e$
- Therefore, the only option that would have a charge of $+1.6 \times 10^{-19} \text{ C}$ is **A**

$$\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$$

B is incorrect because

a quark combination of up, strange, strange would give a charge of

$$-\frac{1}{3}e - \frac{1}{3}e - \frac{1}{3}e = -1e$$

C is incorrect because

a quark combination of down, strange, strange would be a Σ^-

$$-\frac{1}{3}e - \frac{1}{3}e - \frac{1}{3}e = -1e$$

D is incorrect because

a quark combination of down, down, strange would be a Σ^- particle with a charge of -1

5

The correct answer is **B** because:

- A meson contains one quark and one anti-quark bound by a gluon

- The 'up' quark has a charge of $\frac{2}{3}e$
 - Where e is the elementary charge $+1.6 \times 10^{-19} \text{ C}$
- The anti-'down' quark has a charge of $\frac{1}{3}e$
 - $\frac{2}{3}e + \frac{1}{3}e = +1e$
- Therefore, the π^+ particle must have one up quark and one anti-down quark to have an overall charge of $+1e$

- A** is incorrect because a meson with a $u \bar{u}$ combination would be a π^0 particle
- C** is incorrect because a meson with a $d \bar{u}$ combination would be a π^- particle
- D** is incorrect because a meson with a $d \bar{d}$ combination would be a π^0 particle

6

The correct answer is **C** because:

- Nucleus P has a nucleon number $A = 214$ and a proton number $Z = 84$
- It decays via α -emission to form nucleus Q:
 - ${}_{84}^{214}\text{P} \rightarrow {}_{82}^{210}\text{Q} + {}_2^4\alpha$
- Overall, the emission of an α -particle from a nucleus will cause the proton number to decrease by two and the nucleon number to decrease by four, as shown on the diagram
 - So, nucleus Q has a nucleon number $A = 210$ and a proton number $Z = 82$
- It then decays via β -emission to form nucleus R
 - ${}_{82}^{210}\text{Q} \rightarrow {}_{83}^{210}\text{R} + {}_{-1}^0\beta$
- Overall, the emission of a β -particle from a nucleus will cause the proton number to increase by one and the nucleon number to stay the same, as shown on the diagram
 - So, nucleus R has a nucleon number $A = 210$ and a proton number $Z = 83$
- During the decay of Q into R, a neutron is changed into a proton
 - Quark composition of neutron = udd
 - Quark composition of proton = uud

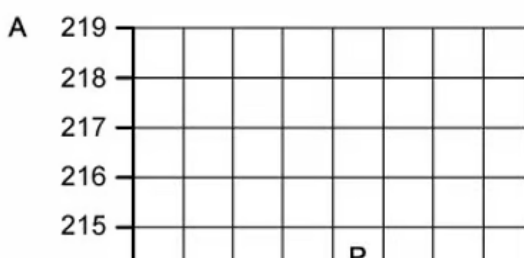
have an overall charge of $+1e$

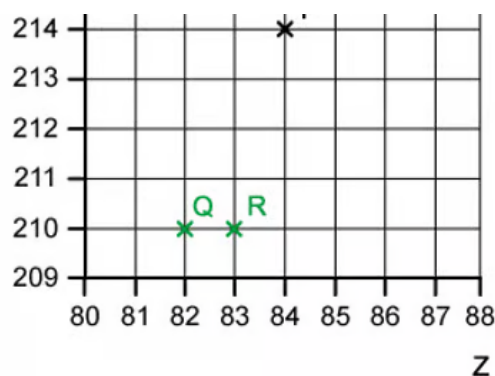
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- During the decay of Q into R, a neutron is changed into a proton
 - Quark composition of neutron = udd
 - Quark composition of proton = uud
- Therefore, a d -quark has changed into a u -quark





7

The correct answer is **B** because

- Protons and neutrons are not antiparticles of each other, instead the weak interaction between quarks gives rise to a change in quark composition
 - Quark composition of neutron = udd
 - Quark composition of proton = uud
- In β^- decay, a neutron turns into a proton ($udd \rightarrow uud$)
- Whereas, in β^+ decay, a proton turns into a neutron ($uud \rightarrow udd$)
- Since electrons and neutrinos are both leptons, they are fundamental particles
- The β^- particle is simply an electron (e^-), and a β^+ particle is its antiparticle – a positron (e^+)
- In β^- decay, an anti-electron neutrino is produced, so, in β^+ decay an electron neutrino is produced
- Therefore, the overall equation for β^+ decay is:
 - $p \rightarrow n + e^+ + \nu_e$

A is incorrect because

in beta-minus and beta-plus decay, a neutron turns into a proton or a proton turns into a neutron due to the weak interaction, their antiparticles are not involved

C is incorrect because

as it is a decay equation, the β^+ particle must be on the right hand side of the reaction

D is incorrect because

in β^- decay, an anti-electron neutrino is produced, so, in β^+ decay, an electron neutrino is produced, an antiparticle is represented by the line through the symbol